Chapter 5: Force and Motion

The following subjects will be discussed

- Newton's Laws of motion.
- Types of Forces
- Examples.

Introduction

Forces in nature. Non-contact forces.

Forces in this course. Contact forces.

From Galileo to Newton. Newtonian motion. Relativistic motion.

Quantum Mechanical dynamics.

Newton's First Law

The "motional" status of an object stays the same if no net external forces is acting on it

Newton's Second Law

If there is a net external force acting on an object, it will accelerate.

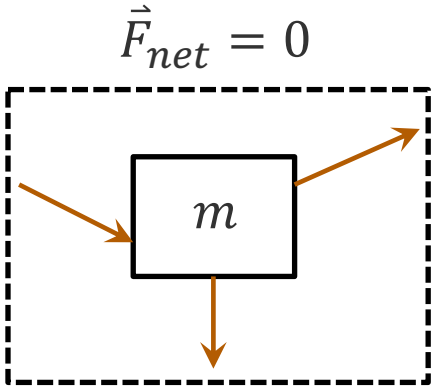
Newton's Third Law

Interacting bodies apply equal (in magnitude) and opposite (in direction) forces on each other

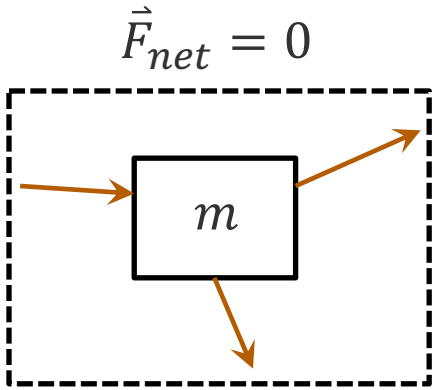
Newton's Laws of Motion

Newton's First Law:

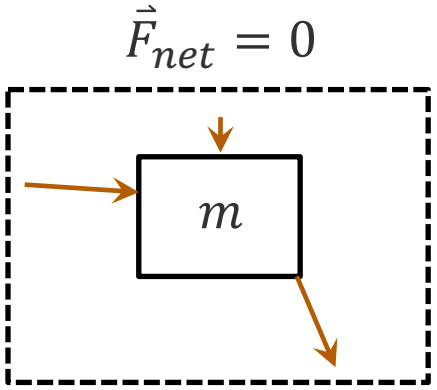
The "motional" status of an object stays the same if no net external forces is acting on it.



Object at rest



Object moves with constant velocity

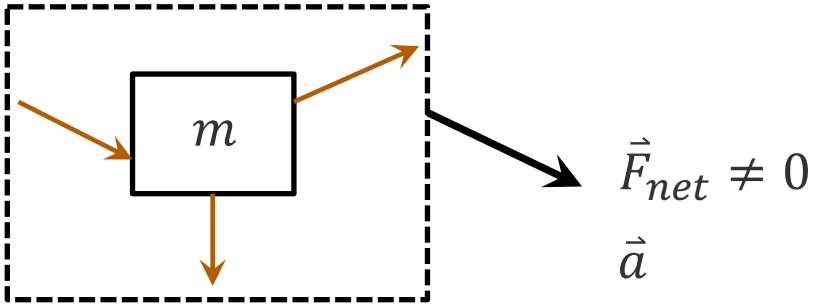


Object moves in a certain direction

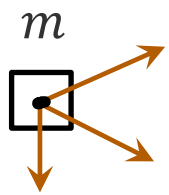
Newton's Laws of Motion

Newton's Second Law:

If there is a net external force acting on an object, it will accelerate.



Object accelerates



The free-body Diagram

1) $\vec{F}_{net} = m\vec{a}$

Units of force is Newton
N ≡ kg·m/s²

2) Acceleration is in the direction as net force.

3) The object is in equilibrium when $\vec{a} = 0$.

4) We can split multi-dimensional motion into

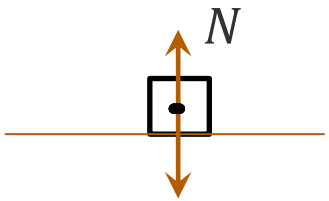
$F_{net_x} = ma_x$
$F_{net_y} = ma_y$
$F_{net_z} = ma_z$

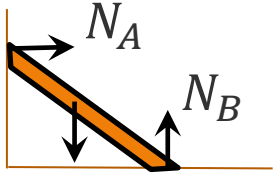
Types of Mechanical Forces

1) Gravitational force. Always directed vertically downward. $F = mg$.

2) The weight of an object is the force it pushes on a flat surface. Not necessarily equal to mg. Example: object on free-fall has zero weight.

3) The normal force (N) is the force that a fixed (not necessarily stationary) surface pushes back at an object resting against it.





4) Friction is the force an object is subjected to when it slides on a rough surface.

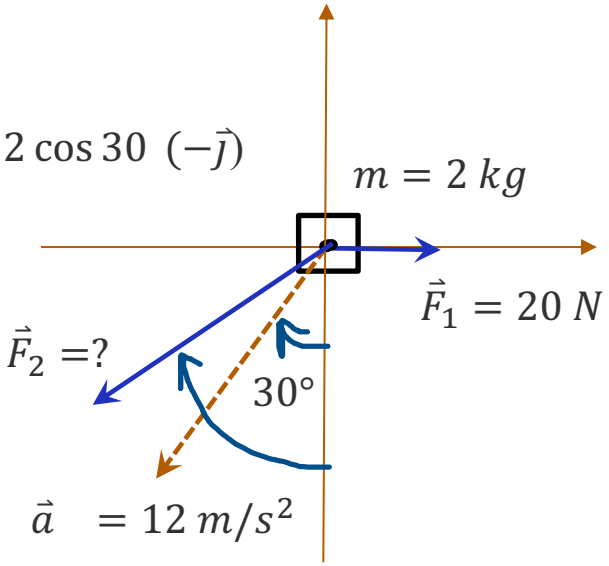
5) Tension is the force created in a connecting wire/rope/cable/cord (a translational force).

6) Drag is the force an object is subjected to when it moves in a viscous medium (fluids/gases).

Examples

1) Find the second force and the angle.

Put in vector form. $\vec{F}_1 = 20\,\vec{\imath}$ $\vec{a} = 12\sin 30\ (-\vec{\imath}) + 12\cos 30\ (-\vec{\jmath})$



$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 = m\,\vec{a} \qquad \vec{F}_2 = m\,\vec{a} - \vec{F}_1$$

$$\vec{F}_2 = (2)(-12\sin 30\ \vec{\imath} - 12\cos 30\ \vec{\jmath}) - 20\,\vec{\imath}$$

$$\vec{F}_2 = -32.0\,\vec{\imath} - 20.8\,\vec{\jmath}\ \ N \qquad angle = \tan^{-1}\frac{32}{20.8} = 57°$$

2) An object of mass 2 kg is subjected to a variable force such that its position at any time is given by $x(t) = 3 + 4t + ct^2 - 2t^3$, where c is a constant. At , $t = 3$ s, $\vec{F}_1 = -36\,\vec{\imath}$ N. Find c.

$$\frac{d^2x}{dt^2} = 2c - 12t$$

$$\left.\frac{d^2x}{dt^2}\right|_{t=3} = 2c - 36$$

At any time, $\vec{F}_{net} = m\,\vec{a}$

At $t = 3$ s, $-36\,\vec{\imath} = (2)\left.\frac{d^2x}{dt^2}\right|_{t=3}\vec{\imath}$ $\rightarrow$ $-36 = (2)(2c - 36)$ $\qquad c = 9\ m/s^2$

Examples

3) Find the tension in the cord.

Object not moving $\quad \vec{F}_{net} = 0$

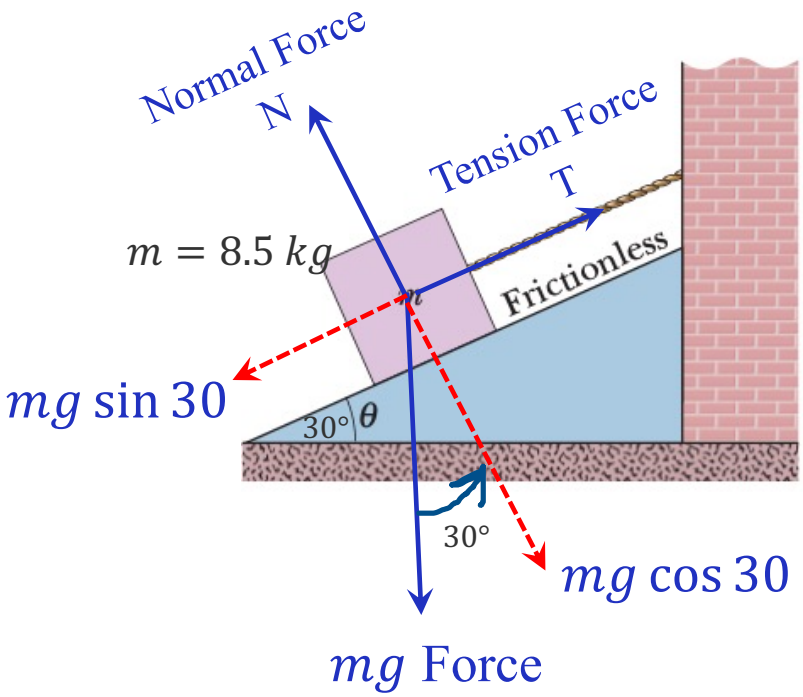
$$T = mg\sin 30 = (8.5)(9.8)(0.5) = 41.7\ N$$

$$N = mg\cos 30 = (8.5)(9.8)\left(\sqrt{3}/2\right) = 72.2\ N$$

The cord is cut, Object will slide with some acceleration.

$$mg\sin 30 = ma$$

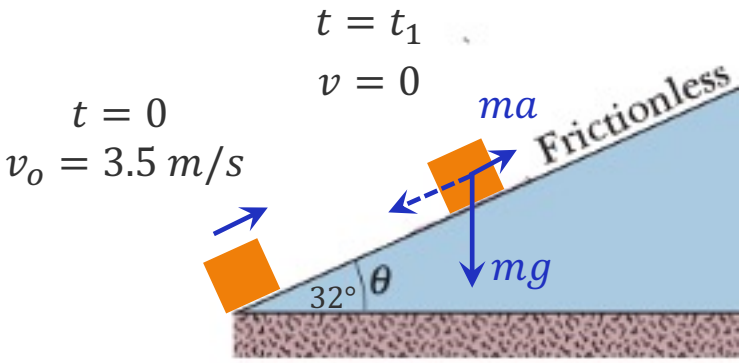
$$a = g\sin 30 = 4.9\ m/s^2$$



Examples

4) Find the distance.

At $t = t_1$, the object stops,
the net force is zero.



$$mg \sin 32 = ma$$

$$a = g \sin 32$$

$$a = -5.2\ m/s^2$$

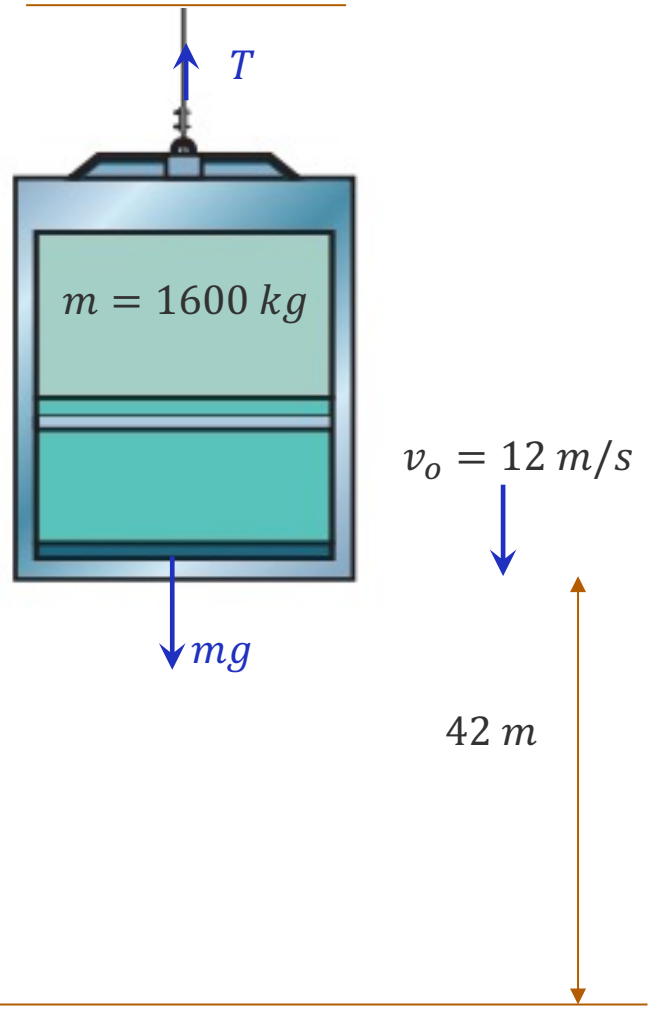
$$v^2 = {v^2}_o + 2a(x - x_o)$$

$$(x - x_o) = -{v^2}_o/2a$$

$$(x - x_o) = 1.18\ m$$

Examples

5) The elevator is brought to the bottom floor (42 m) with constant acceleration. It was initially moving downward with initial velocity 12 m/s. Find the tension in the supporting cable.



$$v^2 = v^2{}_o + 2a(x - x_o)$$

$$a = \frac{-v^2{}_o}{2(x - x_o)}$$

$$a = -1.71\ m/s^2$$

At any time

$$\vec{F}_{net} = m\,\vec{a}$$

$$mg - T = m\,a$$

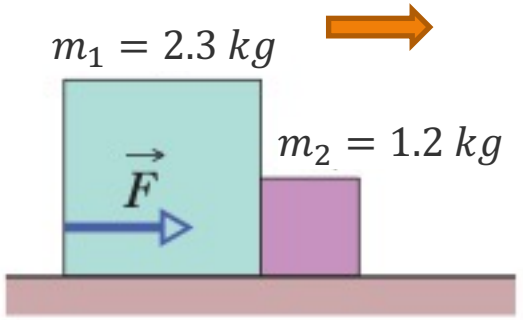
$$T = m(g - a) = (1600)(9.8 + 1.71) = 18.4\ kN$$

Examples

6) The two blocks are in contact and are moved by the force $F = 3.2\ N$. Find the force between the blocks.

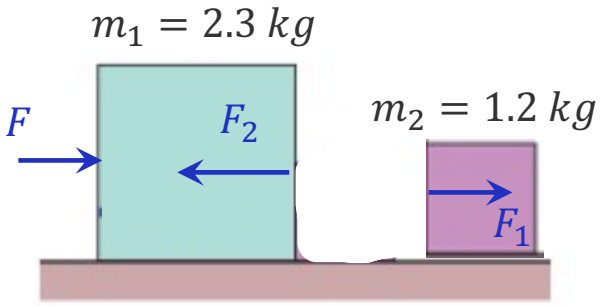
$$F = (m_1 + m_2)a$$

$$F - F_2 = m_1\, a$$



Divide the two equations

$$\frac{F - F_2}{F} = \frac{m_1}{m_1 + m_2}$$



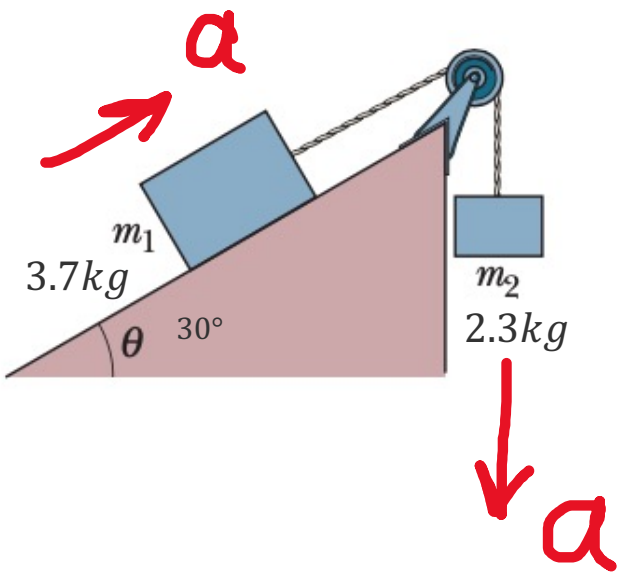
Solve for F_2

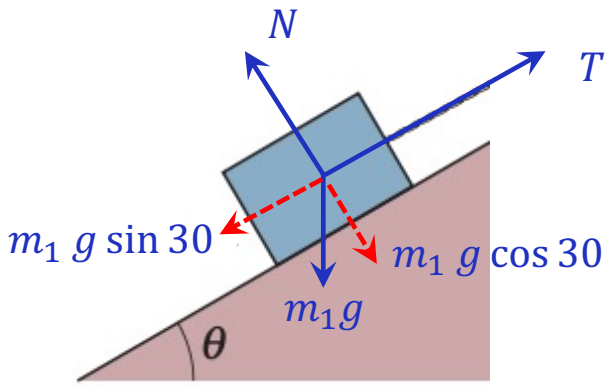
$$F_2 = \frac{m_2}{m_1 + m_2} F = 1.1\ N$$

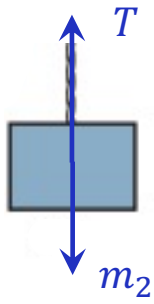
What is F_1?

Examples

7) Find tension and acceleration.







$T - m_1\ g \sin 30 = m_1 a$

$m_2\ g - T = m_2 a$

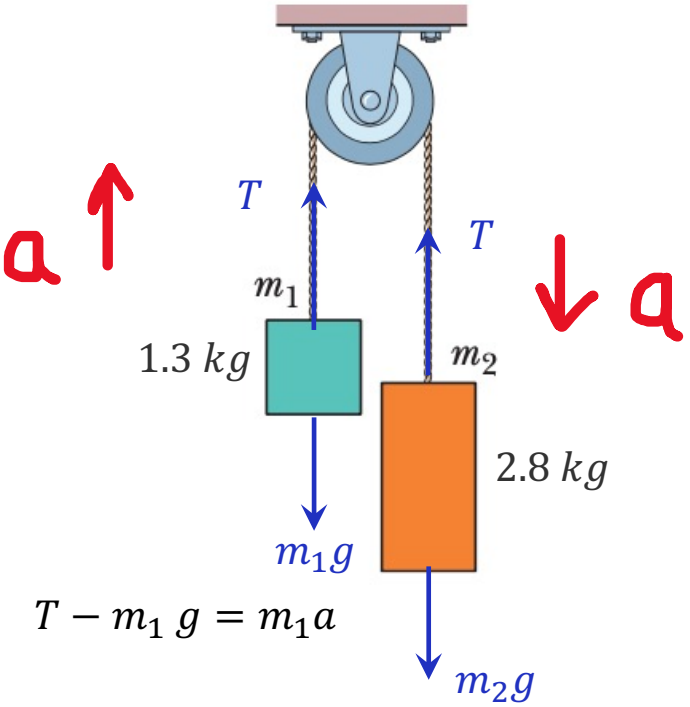
Add the two equations, $m_2\ g - m_1\ g \sin 30 = (m_1 + m_2)a$

$(2.3)\ (9.8) - (3.7)\ (9.8) \sin 30 = (3.7 + 2.3)a$ $\quad a = 0.76\ m/s^2$

Sub in any equation $(2.3)\ (9.8) - T = (2.3)(0.76)$ $\quad T = 20.8\ N$

Examples

8) Find tension and acceleration.



$T - m_1\, g = m_1 a$

$m_2\, g - T = m_2 a$

$a = 3.6\ m/s^2$

$T = 17.4\ N$

Bag 1 is now ($t = 0$) losing some sand at 0.2 kg/s. Find the rate of change in acceleration at $t = 0$ and $t = 3\ s$.

$T - m_1\, g = m_1 a$ $\qquad \frac{dT}{dt} - \frac{dm_1}{dt} g = \frac{dm_1}{dt} a + m_1 \frac{da}{dt}$ ✓

$m_2\, g - T = m_2 a$ $\qquad 0 - \frac{dT}{dt} = 0 + m_2 \frac{da}{dt}$ ✓

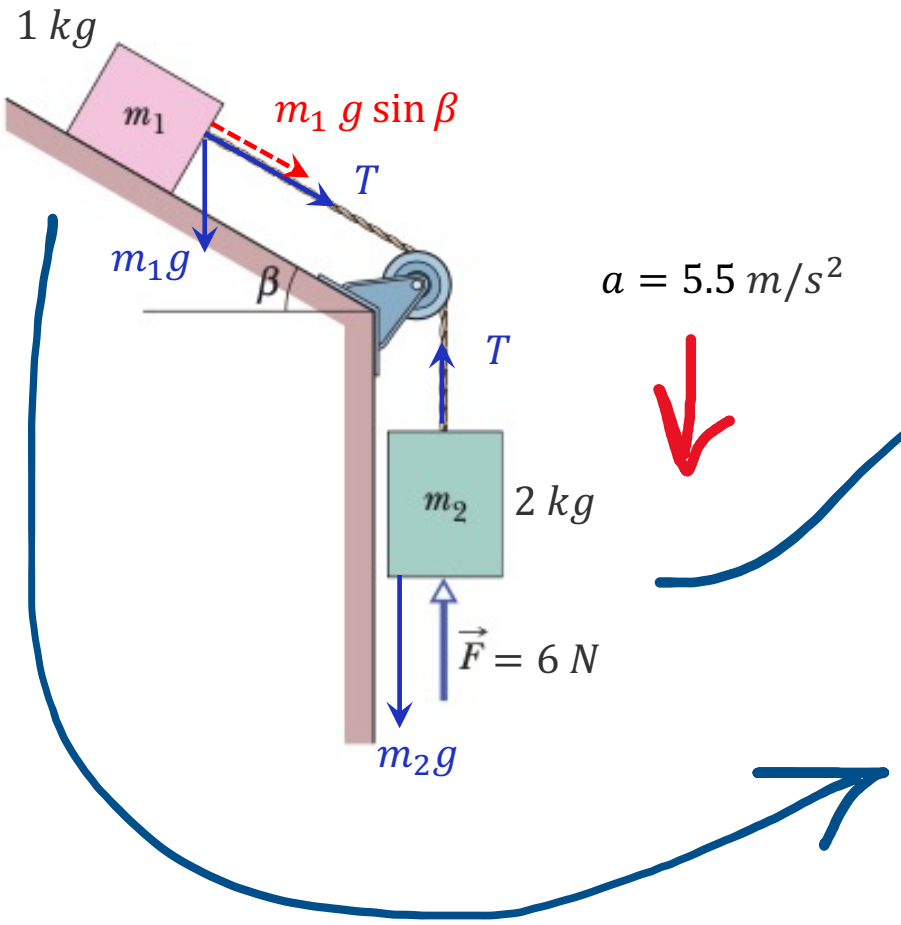
Add the two equations, and solve for $\frac{da}{dt}$

$$\frac{da}{dt} = \frac{\frac{dm_1}{dt}(g + a)}{-(m_1 + m_2)}$$

$t =$	0	1	2	3
m_1				
a				
$\frac{da}{dt}$				

Examples

9) Find tension and angle.



$$m_2\ g - (T + F) = m_2 a$$

$$T = m_2\ g - F - m_2 a$$

$$T = 2.6\ N$$

$$a = 5.5\ m/s^2$$

$$T + m_1\ g \sin\beta = m_1 a$$

$$\sin\beta = \frac{m_1 a - T}{m_1 g}$$

$$\mathrm{B} = 17.2°$$

Q & A

Send me an email if you have any questions.